# 5 Layouts

***This chapter covers***

- Histogram and pie chart layouts
- Simple tweening
- Tree, circle pack, and stack layouts
- Sankey diagrams and word clouds

D3 contains a variety of functions, referred to as *layouts*, that help you format your data so that it can be presented using a popular charting method. In this chapter we'll look at several different layouts so that you can understand general layout functionality, learn how to deal with D3's layout structure, and deploy one of these layouts (some of which are shown in figure 5.1) with your data.

In each case, as you'll see with the following examples, when a dataset is associated with a layout, each of the objects in the dataset has attributes that allow for drawing the data. Layouts don't draw the data, nor are they called like components or referred to in the drawing code like generators. Rather, they're a preprocessing step that formats your data so that it's ready to be displayed in the form you've chosen. You can update a layout, and then if you rebind that altered data to your graphical objects, you can use the D3 enter/update/exit syntax you encountered in chapter 2 to update your layout. Paired with animated transitions, this can provide you with the framework for an interactive, dynamic chart.